b. Which job has the highest salaries? Illustrate with bar graph.

```
In [31]: # Job with the highest salary

         df.loc[df['salary_in_usd'] == df['salary_in_usd'].max()]
```

Out[31]:

	work_year	experience_level	employment_type	job_title	salary_in_usd	employee_residence	remote_ratio	company_location	company_size
3522	2020	Medium Level/Intermediate	FT	Research Scientist	450000	US	0	US	M

```
In [32]: # Top 10 jobs with highest maximum salaries

         top_10_highest_salary_jobs = df.groupby('job_title')['salary_in_usd'].max().sort_values(ascending=False).head(10)
         top_10_highest_salary_jobs
```

Out[32]:

```
job_title
Research Scientist                    450000
Data Analyst                          430967
AI Scientist                          423834
Applied Machine Learning Scientist    423000
Principal Data Scientist              416000
Data Scientist                        412000
Data Analytics Lead                   405000
Applied Data Scientist                380000
Data Architect                        376080
Data Science Tech Lead                375000
Name: salary_in_usd, dtype: int64
```

```
In [20]: # Showing top 10 jobs with highest salaries in bar graph

         plt.bar(top_10_highest_salary_jobs.index, top_10_highest_salary_jobs.values,
                 color = ['#1f77b4', '#ff7f0e', '#2ca02c', '#d62728', '#9467bd',
                    '#8c564b', '#e377c2', '#7f7f7f', '#bcbd22', '#17becf'])

         plt.title("Top 10 Jobs", fontsize=17)
         plt.xlabel("Position")
         plt.ylabel ("Average Salary ($)")
         plt.xticks(rotation=90)

         plt.show()
```

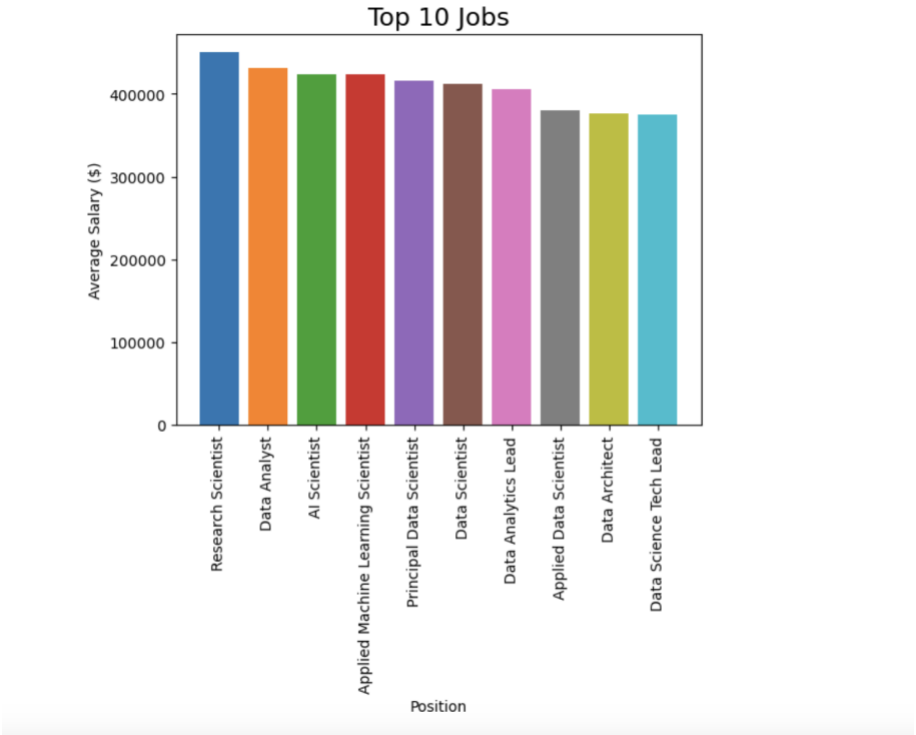


Figure 17: Screenshot of highest salary jobs

Using the groupby function, job titles are grouped according to their maximum salary in USD. The 'sort_values' method is used, and the ascending parameter is set to false to arrange them in descending order. Finally, the 'head(10)' method is used to retrieve the top 10 jobs with the highest salaries. This pandas series is stored in the 'top_10_highest_salary_jobs' variable.

Now, the index and values, i.e. jobs and their maximum salaries, are passed as parameters to the 'bar' method of matplotlib's pyplot library to plot a bar graph. Respective methods are used for title, xlabel, ylabel, and finally, the bar graph is displayed using the 'show' method.

From the above results, we can conclude that research scientist is the job with highest salary i.e. 450000 USD$. It is followed by data analyst, AI scientist and others. However, there is not much difference in the highest salaries of these jobs.

c. Write a python program to find out salaries based on experience level. Illustrate it through bar graph.

```
In [21]: # Average salary based on experience

         exp_avg_salary = df.groupby('experience_level')['salary_in_usd'].mean()
         exp_avg_salary
```

```
Out[21]: experience_level
         Entry Level                  72648.685185
         Execute level               191078.208333
         Medium Level/Intermediate   101828.783133
         Senior Level/Expert         153897.435650
         Name: salary_in_usd, dtype: float64
```

```
In [22]: # Showing average salary based on experience salaries in bar graph

         plt.bar(exp_avg_salary.index, exp_avg_salary.values,
                 color=['#1f77b4', '#ff7f0e', '#2ca02c', '#d62728'])

         plt.title("Salaries Based On Experience", fontsize=17)
         plt.xlabel("Experience Level")
         plt.ylabel("Average Salary ($)")
         plt.xticks(rotation=15)
         plt.grid(axis='y', linestyle='--', alpha=0.7)

         plt.show()
```

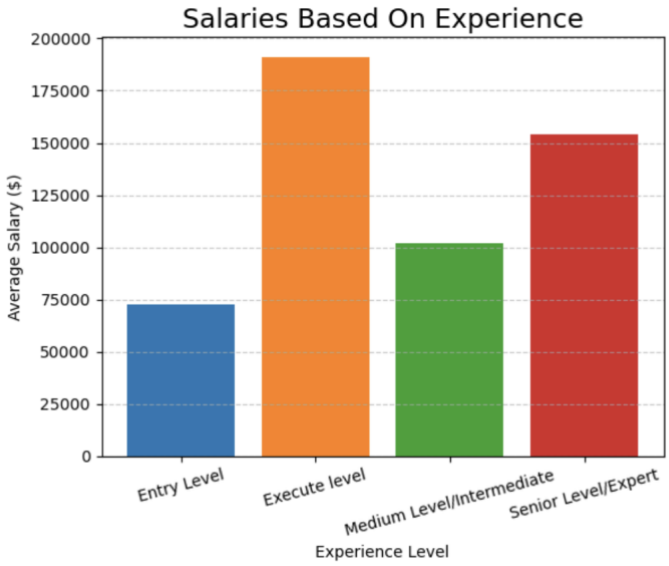


Figure 18: Screenshot of average salary based on experience

Here, the groupby function is used to group the 'experience_level' using mean as the aggregate function. This series is stored int exp_avg_salary variable. Again, the bar method is called passing index and of the above series as parameter to plot the bar graph. The title, labels, colors, and other attributes are given by using their respective functions. Finally, "show" function displays the bar graph.

From the above graph, we can say that higher the position, higher the salary. Execute level have the highest average salary due to their extensive experiences, leadership responsibilities, and strategic roles in the organization. It is followed by senior/ expert level as they are particularly specialized in a field, more experience, and advanced qualifications. Then comes medium level/ intermediate with average salary of about 100,000 USD$ and then finally entry level experience with about 72,000 USD$ average salary.

d. Write a Python program to show histogram and box plot of any chosen different variables. Use proper labels in the graph.

```
In [24]: # Histogram

         plt.figure(figsize=(10, 6))

         bins = int((df['salary_in_usd'].max() - df['salary_in_usd'].min()) / 20000)
         plt.hist(df['salary_in_usd'], bins=bins, color='skyblue', edgecolor='black')
         plt.title("Histogram", fontsize=17)
         plt.xlabel("Salary in USD")
         plt.ylabel("Frequency")

         plt.show()
```

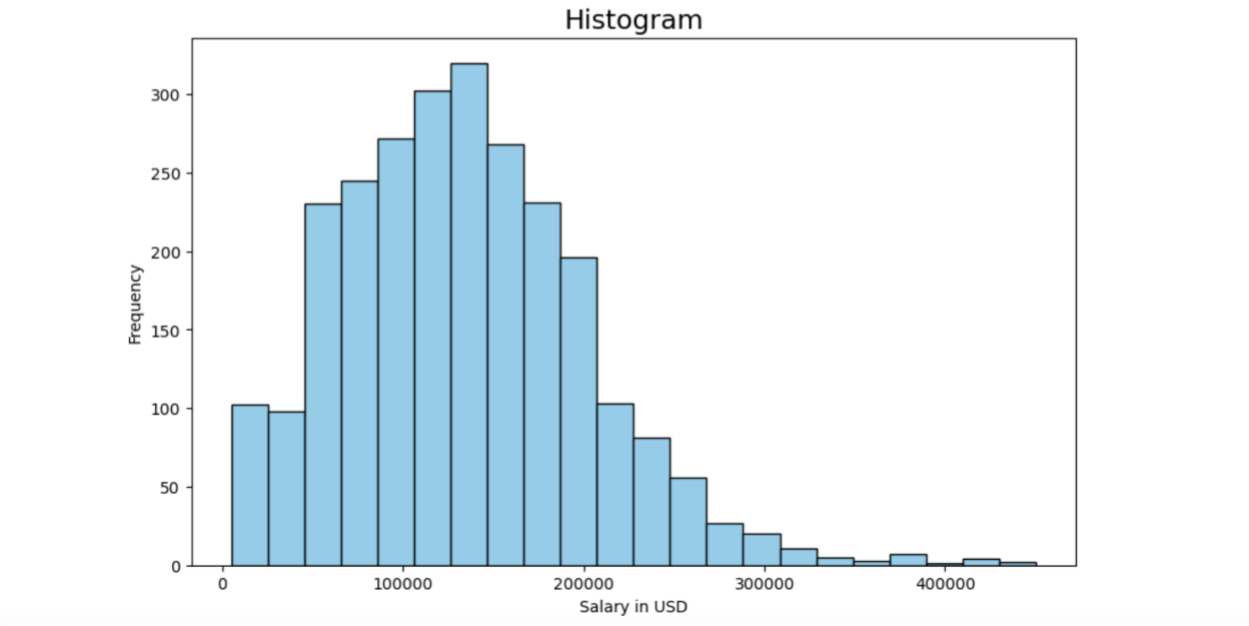


Figure 19: Screenshot of histogram of salary in usd

The plt.figure() function called at the top with figure size 10 by 6 given at parameters determines the dimensions of the figure of the graph. Range of salary_in_usd column is divided by 20000(bin width) , its integer value is set to bins variable. It determines the number of bins. Now, hist function of plt library is called with respective parameters to plot the historgram of salary_in_usd column. The title, labels, colors, and other attributes are given by using their respective functions. Finally, “show” function displays the histogram.

According to this histogram, the frequency of salary between 100000 to 200000 is the highest. It also shows that the distribution is right skewed which means that there are relatively more jobs with lower salaries and a few jobs with very high salaries.

```
In [25]: # Box Plot

plt.boxplot(df['salary_in_usd'], patch_artist=True, boxprops=dict(facecolor='brown'))
plt.title("Box Plot", fontsize=17)
plt.xlabel ("Salary in USD")

plt.show()
```

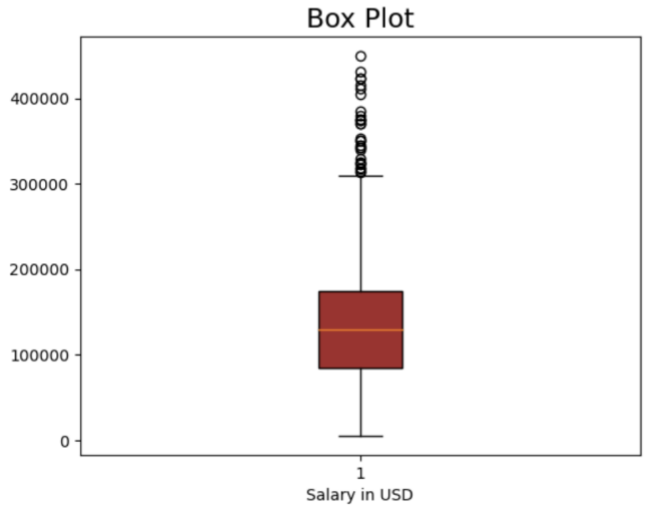


Figure 20: Screenshot of boxplot of salary in usd

Similarly, the boxplot function of matplotlib library is used for plotting the boxplot. Passing the column (salary_in_usd in our case), patch_artist and boxprops for color, creates our boxplot. The title and xlabel are given respectively and finally plt.show displays the boxplot for salary_in_usd.

In the displayed boxplot, we can see that the median salary in usd lies between 100000 and 200000 USD$. Similarly, the upper quartile also lies below 200000 USD$ and the lower quartile is below 100000 USD$. The outliers shown in the boxplot are the jobs with exceptionally high salary.

6. Conclusion

A thorough evaluation of the python notebook has been conducted in this report, and all the findings have been found to be satisfactory. All the data wrangling that was carried out were successful, and all the project's objectives outlined in the coursework were met, with all the errors and bugs found being fixed.

This coursework has been a valuable learning experience, helping me gain knowledge, skills, and experience. I became much more familiar with pandas and matplotlib, and other tools such as Anaconda-Navigator, Jupyter Notebook, MS Word, and many more. This real-world scenario problem also deepened my understanding of data analytics is done. Additionally, this project enhanced my research and report writing skills. All these skills and knowledge will be useful in my future endeavors.

Many challenges were encountered and solved during my coursework journey. I had to seek help from my teachers and friends to solve the problems faced. Similarly, I also faced difficulties during the report, such as formatting it correctly, crashing of MS Word and Eclipse IDE. I had to rely on Google for solutions and learned the importance of making backups after completing a major portion of the program and report.

Finally, this project has been a challenging but rewarding experience. I am proud that I was able to complete this project, and I would like to thank everyone, especially all my teachers, who helped me make this project a success. It was a great learning experience, and I am grateful to have been a part of it.